

Trainings ▾

Preparatory Steps

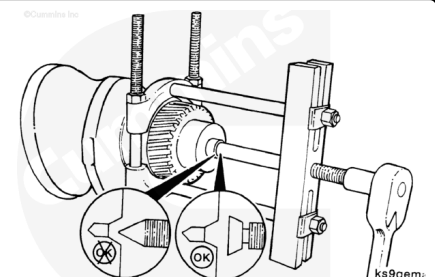
- Remove the crankshaft. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-016.html>)



Remove

⚠ CAUTION ⚠

Do not try to split the front crankshaft gear to remove. The gear is made out of steel and will not split. Damage to the tool and the crankshaft can result.



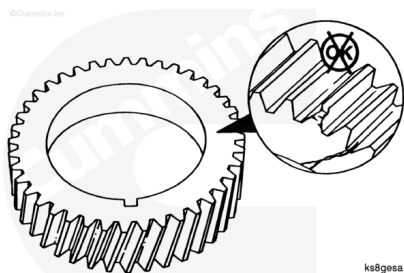
Note : If a front crankshaft seal wear sleeve has been installed during a previous repair, it **must** be removed before the crankshaft gear is removed. Refer to Procedure 001-025 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-025.html>)

Use a heavy-duty bearing separator, Part Number 316427, or gear puller, as illustrated, to remove the crankshaft gear.

Clean and Inspect for Reuse

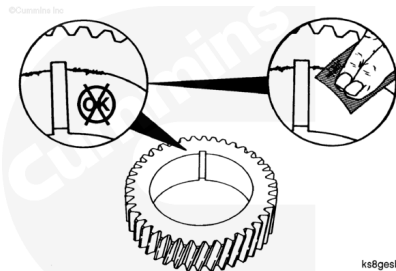
Inspect for cracks, broken, or chipped teeth.

The gear **must** be replaced if it is damaged.



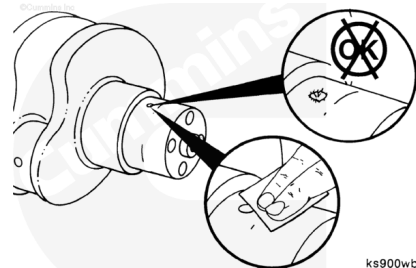
Inspect the gear and keyway for nicks or burrs.

Use fine crocus cloth to remove nicks and burrs.



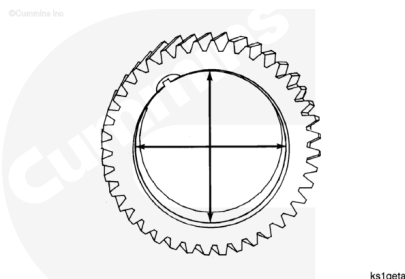
Inspect the crankshaft gear journal and the gear alignment dowel pin hole for burrs or damage.

Use fine crocus cloth to remove burrs.



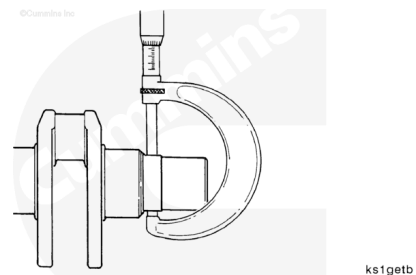
Measure the crankshaft gear bore inside diameter.

Crankshaft Gear Bore Inside Diameter		
mm		in
75.898	MIN	2.9881
75.923	MAX	2.9891



Measure the crankshaft gear journal outside diameter. Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/41/41-001-016.html)



Install

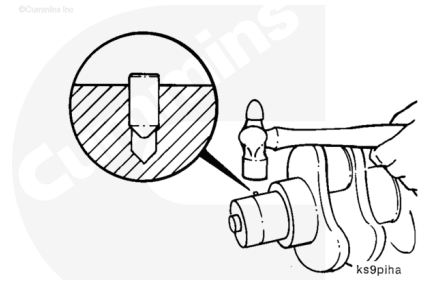
⚠ CAUTION ⚠

Do not exceed the specified heating time or temperature. The crankshaft teeth can be damaged.

Use a soft hammer to install the gear alignment dowel into the crankshaft.

The pin **must** be 1.02 mm [0.040 in] above the crankshaft surface.

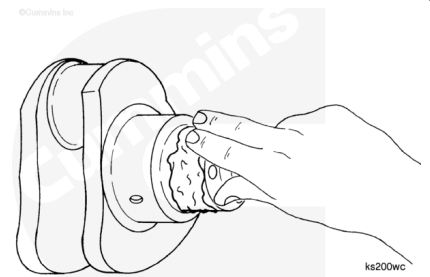
Heat the gear in an oven for a minimum of 45 minutes, but **not** more than 2 hours.



Measurements

		celsius	fahrenheit
Temperature	Steel Gear	177	350

Use assembly lubricant, Part Number 3163087 or equivalent, to lubricate the outside diameter of the crankshaft gear journal.

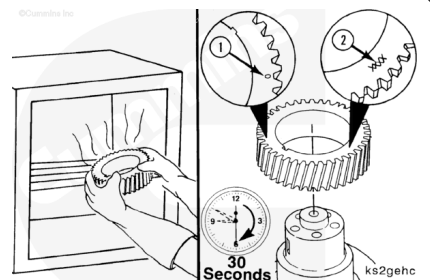


⚠ WARNING ⚠

Wear protective gloves to reduce the possibility of personal injury when handling parts that have been heated.

⚠ CAUTION ⚠

The timing mark (1) and part number (2) on the gear must be facing away from the crankshaft after the gear is installed. Engine damage can result if the gear is installed backwards.



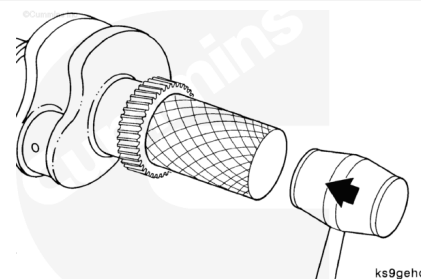
⚠ CAUTION ⚠

Do not use water or oil to reduce the cooling time. The gear can crack. Allow the gear to air cool.

Remove the gear from the oven.

Align the keyway of the gear with the alignment dowel pin in the crankshaft and install the gear within 30 seconds.

If the gear cools and stops on the crankshaft before it is fully installed, use a driver to complete the installation.



Finishing Steps

- Install the crankshaft. Refer to Procedure 001-016 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-016.html>)
- Operate the engine and check for leaks.

